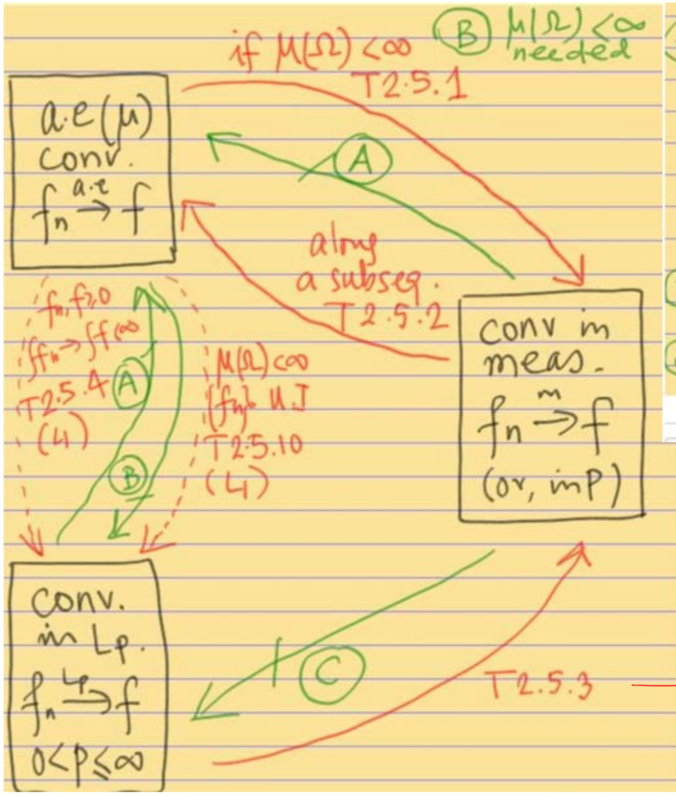


→ In fact if $f_n \xrightarrow{L_\infty} f$ then $f_n \to f$ a.e(μ)

(A) moving indicators:

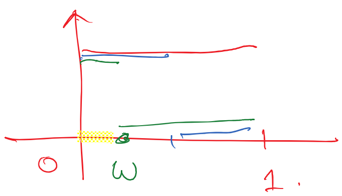
$f_n(x) = 1_{\left[\frac{i}{2^k}, \frac{i+1}{2^k}\right]}$ where $\boxed{n = 2^k + i}$, $i = 0, \dots, 2^k - 1$

$(\Omega, \mathcal{F}) = ([0,1], \mathcal{B}[0,1] = [0,1] \cap \mathcal{B}(\mathbb{R}))$, $f \equiv 0$

(B) infinite tail indicators, $f_n = 1_{[n,\infty)}$, $(\Omega, \mathcal{F}) = (\mathbb{R}_+, \mathcal{B}(\mathbb{R}_+))$, $f \equiv 0$

(C) tall-thin indicators: $f_n = n 1_{[0, \frac{1}{n}]}$, $(\Omega, \mathcal{F})$, f as in (A)

(A)



$f_1 = 1_{[0,1]}$ $n = 1 = 2^0 + 0$ $k = 0$

$f_2 = 1_{[0, \frac{1}{2})}$, $f_3 = 1_{[\frac{1}{2}, 1]}$ $\underline{k = 1}$

$2 = 2^1 + 0$, $3 = 2^1 + 1$

$f_4 = 1_{[0, \frac{1}{4}]}$, $f_5 = 1_{[\frac{1}{4}, \frac{1}{2}]}$, $f_6 = 1_{[\frac{1}{2}, \frac{3}{4}]}$, $f_7 = 1_{[\frac{3}{4}, 1]}$ $k = 2$

$5 = 2^2 + 1$

$k = 3$

fix $\omega \in \Omega = [0,1]$

$f_n(\omega) = 0$ or 1 infinitely many times

$f \equiv 0$ $0 < \varepsilon < 1$

$$\mu(\{\omega : |f_n(\omega) - \underset{0}{f(\omega)}| > \varepsilon\}) = \frac{1}{2^k} \text{ if } \underline{2^k \le n < 2^{k+1}}$$

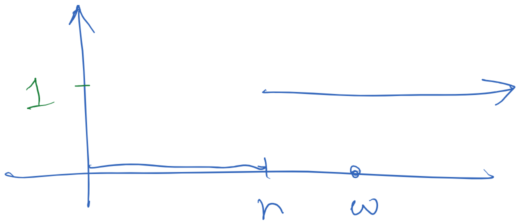
$$\frac{1}{2^k} = \frac{2}{2^{k+1}} < \frac{2}{n}.$$

So $f_n \xrightarrow{\text{in meas}} f$ but $f_n \overset{a.e}{\not\to} f$ $< \frac{2}{n} \to 0$ as $n \to \infty$

(B) $f_n = 1_{[n,\infty)}$, $(\Omega, \mathcal{F}, \mu) = (\mathbb{R}_+, \mathcal{B}(\mathbb{R}_+), m)$

$\mathbb{R}_+ = [0, \infty)$

$f \equiv 0$



$f \equiv 0$

$f_n \xrightarrow{a.e(\mu)} f$ but not in meas.

C

$\leftarrow f_n$.

$f_n \xrightarrow{a.e(\mu)} f$ but not in L_1

* check the other statements in diagram.

take $\varepsilon > 0$. if $\varepsilon \geq 1$ Then

$$\mu(|f_n - \overset{\equiv 0}{f}| > \varepsilon) = 0$$

if $0 < \varepsilon < 1$

$$\mu(|f_n - f| > \varepsilon) = \mu((n, \infty)) = \infty$$

$\not\to 0$ as $n \to \infty$.

for $\omega \in \Omega = \mathbb{R}_+$

take $N > \omega$, $f_n(\omega) = 0 \quad \forall n > N$.

$\to f(\omega) = 0$

Thm 2.5.1: $\mu(\Omega) < \infty$,

given $\mu(\lim_{n\to\infty} f_n \neq f) = 0$, show $\forall \varepsilon > 0$ $\lim_{n \to \infty} \mu(|f_n - f| > \varepsilon) = 0$

$\Updownarrow$

$$\mu(\underbrace{\limsup_{n\to\infty} |f_n - f| > 0}_{A}) = 0$$

$\Uparrow$

$\Longrightarrow$ will show $\forall r \geq 1$

$$\lim_{n\to\infty} \mu(|f_n - f| \geq \tfrac{1}{r}) = 0$$

take any $\varepsilon > 0$, choose

$r > \frac{1}{\varepsilon}$ then $\underline{\frac{1}{r} \leq \varepsilon}$

$$\mu\left(\underbrace{|f_n - f| > \varepsilon}\right) \leq \mu\left(|f_n - f| \geq \frac{1}{r}\right) \to 0$$

$$A = \left\{ \omega : \limsup_{n\to\infty} |f_n(\omega) - f(\omega)| > 0 \right\}.$$

$$= \left\{ \omega : \underline{\limsup} \, |f_n(\omega) - f(\omega)| \geq \frac{1}{r} \text{ } \underline{\text{for some}} \, r \geq 1 \right\}$$

$$= \bigcup_{r \geq 1} \left\{ \omega : \underline{\inf_{n \geq 1} \left(\sup_{j \geq n} \right.} |f_j(\omega) - f(\omega)| \left. \right) \geq \frac{1}{r} \right\}.$$

$$= \bigcup_{r \geq 1} \left\{ \omega : \sup_{j \geq n} |f_j(\omega) - f(\omega)| \geq \frac{1}{r}, \ \underline{\underline{\forall \, n \geq 1}} \right\}$$

$$= \bigcup_{r \geq 1} \bigcap_{n \geq 1} \left\{ \omega : \underline{\underline{\sup_{j \geq n}}} |f_j(\omega) - f(\omega)| \geq \frac{1}{r} \right\}$$

$$= \bigcup_{r \geq 1} \bigcap_{n \geq 1} \bigcup_{j \geq n} \underbrace{\left\{ \omega : |f_j(\omega) - f(\omega)| \geq \frac{1}{r} \right\}}_{B_j(r)}.$$

$$C_n(r) = \bigcup_{j \geq n} \underline{B_j(r)}$$

$$D(r) = \bigcap_{n \geq 1} C_n(r)$$

$$A = \bigcup_{r \geq 1} D(r) \quad \text{and} \quad \mu(A) = 0$$

$$\mu(A) = 0 \implies \mu(D(r)) \leq \mu(A) = 0 \quad \forall \, r \geq 1$$

$C_n(r) \downarrow D(r)$ as $n \to \infty$. and $\mu(\Omega) < \infty$

so mcfa applies : $\lim_{n\to\infty} \mu(C_n(r)) = \mu(D(r)) = 0$

Note $\mu(B_j(r)) \leq \mu(C_n(r)) \quad \forall j \geq n$.

$\mu(B_n(r)) \leq \mu(C_n(r)) \to 0$ as $n \to \infty \quad \forall r \geq 1$

$$\lim_{n\to\infty} \mu(B_n(r)) = 0 \qquad \forall r \geq 1$$

i.e, $\lim_{n\to\infty} \mu\left(\left\{\omega : |f_n(\omega) - f(\omega)| \geq \frac{1}{r}\right\}\right) = 0$

pf complete.